

Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Powerline Networking System.

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

A.	Full name	ABC
	Mobile Number	805371XXXX
	Email (personal)	abc@gmail.com
	UID/Registration number	XXXXXX
	Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
	Signature (Mandatory)	Kindly provide the signature.
B.	Full name	DEF
	Mobile Number	9782801XXXX
	Email (personal)	def@gmail.com
	UID/Registration number	XXXXXX
	Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
	Signature (Mandatory)	Kindly provide the signature.
C.	Full name	GHI
	Mobile Number	702845XXXX
	Email (personal)	ghi@gmail.com

UID/Registration number	XXXXXX
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
Signature (Mandatory)	Kindly provide the signature.

EXTERNAL INVENTOR(S): (INVENTORS NOT WORKING IN LPU)

A.	Full name	Provide relevant details
	Mobile Number	
	Email	
	Address of External Affiliations	
	Signature (Mandatory)	

For External Inventors, NOC (No Objection Certificate) from the affiliated institute/university/Industry/lab etc. is mandatory for each individual inventor and their respective topic. For NOC, format is attached below.

3. DESCRIPTION OF THE INVENTION: The powerline is able to carry data via the superposition of a low-energy information signal to the power wave, which means they can pass through each other without being disturbed. Data is transmitted at a minimum of 3 kHz to ensure that the power wave does not interfere with the data signal. Electrical wiring in the home is able to transmit signals at a variety of frequencies. Electricity usually travels at 50/60 Hz frequencies, meaning that data can use the same wires, but at a much higher frequency so that the two do not affect each other. In industrial and commercial settings, powerline networking has various uses. It has been used in the smart grid for advanced energy management, fraud detection, automatic meter readings, street lighting control, and remote metering and billing.

A. PROBLEM ADDRESSED BY THE INVENTION: In a residential area, a powerline network is implemented to transmit data alongside the conventional power wave. The data signal operates at a minimum frequency of 3 kHz to ensure interference-free coexistence with the 50/60 Hz power wave. Explain the significance of using a higher frequency for data transmission and how it allows the two signals to share the same electrical wiring without disruption. Additionally, discuss the versatility of electrical wiring in homes to carry signals at various frequencies. In the context of industrial and commercial settings,

elaborate on the diverse applications of powerline networking, such as smart grid functionalities, advanced energy management, fraud detection, automatic meter readings, street lighting control, and remote metering and billing.

B. OBJECTIVE OF THE INVENTION (Provide minimum two)

Objective 1: Evaluate home powerline networking efficiency at a minimum 3 kHz frequency for data transfer alongside the 50/60 Hz power wave. Evaluate the ability of residential electrical wiring to tolerate signals at multiple frequencies, as well as the consequences for the harmonious coexistence of power and data transmission.

Objective 2: Explore the many uses of powerline networking in industrial and commercial contexts, with an emphasis on its function in smart grids. Investigate the role of this technology in sophisticated energy management, fraud detection, automatic metre readings, street lighting control and remote metering and invoicing. Evaluate the possible benefits and drawbacks of deploying powerline networking in various urban and industrial scenarios.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY: Describe your invention fulfil the research gap?

Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
1.	CN105577236A	The invention relates to a power system communication networking system comprising a PLC system transmitting end and a receiving end. The PLC system transmitting end is provided with a serial/parallel converter, a base band modulation module, an IFFT, a D/A converter, and a combiner, which are sequentially connected together. The receiving end is provided with an A/D converter, a band pass filter, a FFT, a demodulation module, a parallel/serial converter,	It emphasizes on a combination of data modulation techniques like OFDM and chaotic modulation for high data rates and security. Whereas, the prior art focused on the technical components of a power system communication networking system,	In our invention we use OFDM and chaotic modulation.

		which are sequentially connected together. The system also comprises an INT 5500chip group, which comprises an INT 5500 base band processor and an INT1200 analog front-end chip. The internal structure is mainly constituted by an I/O interface function module, a Power Packet Mac/PHY module, and an ADC/DAC module. The power system communication networking system is advantageous in that for the broadband power access network, the data acquisition can be effectively carried out, the transmitting and the processing of various data can be timely carried out; by using the high-speed interface, the data transfer and the data acquisition processing	describing the sequential connections and internal structure of the system.	
2.	US20060018328A1	A method and system for powerline networking. The method and system include an adapter and a gateway connected to powerlines. The gateway is operable for configuring the adapter for powerline networking by communicating configuration signals to the adapter over the powerlines. A powerline network is defined by the configuring of the adapter to communicate with the gateway over the powerlines.	The prior art provides a concise description of a method and system for powerline networking, emphasizing the interaction between an adapter and a gateway over powerlines to establish a powerline	We use the technical aspects, modulation techniques, or applications

			network. It does not delve into technical aspects, modulation techniques, or applications like the first text did. Instead, it focuses on the process of configuring devices for powerline networking using powerlines as the communication medium.	
3.	Anatory, Justinian, Nelson Theethayi, and Rajeev Thottappillil. "Effects of multipath on OFDM systems for indoor broadband power-line communication networks." IEEE Transactions on power delivery 24.3 (2009): 1190-1197.	Power-line networks are an excellent infrastructure for broadband data transmission. However, various multipaths within a broadband power-line communication (BPLC) system exist due to stochastic changes in the network load impedances, branches, etc. This further affects network performance. This paper attempts to investigate the performance of indoor channels of a BPLC system that uses orthogonal frequency-division multiplexing (OFDM) techniques. It is observed that when a branch is added in the link between the sending and receiving end of an indoor channel, an average of 4-dB power loss is found. Additionally, when the	The authors investigation of the performance of indoor channels in a broadband power-line communication (BPLC) system using orthogonal frequency-division multiplexing (OFDM) techniques. Changes in the terminal impedances of the branch also affect the power loss, with a decrease in impedance leading to a 0.67 dB/Omega loss,	A power-line networking system that combines OFDM and chaotic modulation for clean energy transmission, aligning with broader sustainable development goals.

		terminal impedances of the branch change from the line characteristic impedance to impedance of lower values, the power loss (signal-to-noise ratio) is about 0.67 dB/Omega. On the contrary, for every increase in the terminal impedances by 100 Omega, above the line characteristic impedance, the power loss is 0.1 dB/Omega. When the line terminal impedances are close to short or open circuits, OFDM techniques show degraded performance. This situation is also observed when the number of branches increases. In this paper, it is shown that to overcome such performance degradation, the concatenated Reed-Solomon codes/interleaved Viterbi methods can be used. The observations presented in the paper could be useful for an efficient design of a BPLC system that uses OFDM techniques.	and an increase by 100 Omega above the line characteristic impedance resulting in a 0.1 dB/Omega loss. Where in our invention, the power-line networking system that utilizes both orthogonal frequency division multiplexing (OFDM) and chaotic modulation for data transmission over electrical wiring. The goal is to contribute to affordable and clean energy, aligning with the broader aim of ensuring access to affordable, reliable, sustainable, and modern energy for all. The system includes a transmitter for modulating data and a receiver for demodulating the data, both using a	
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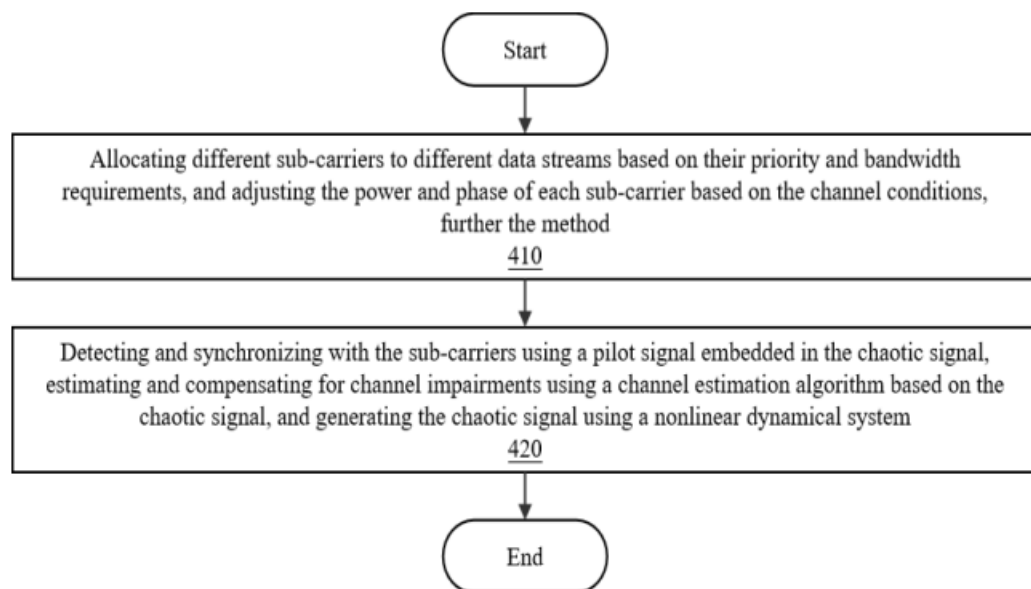
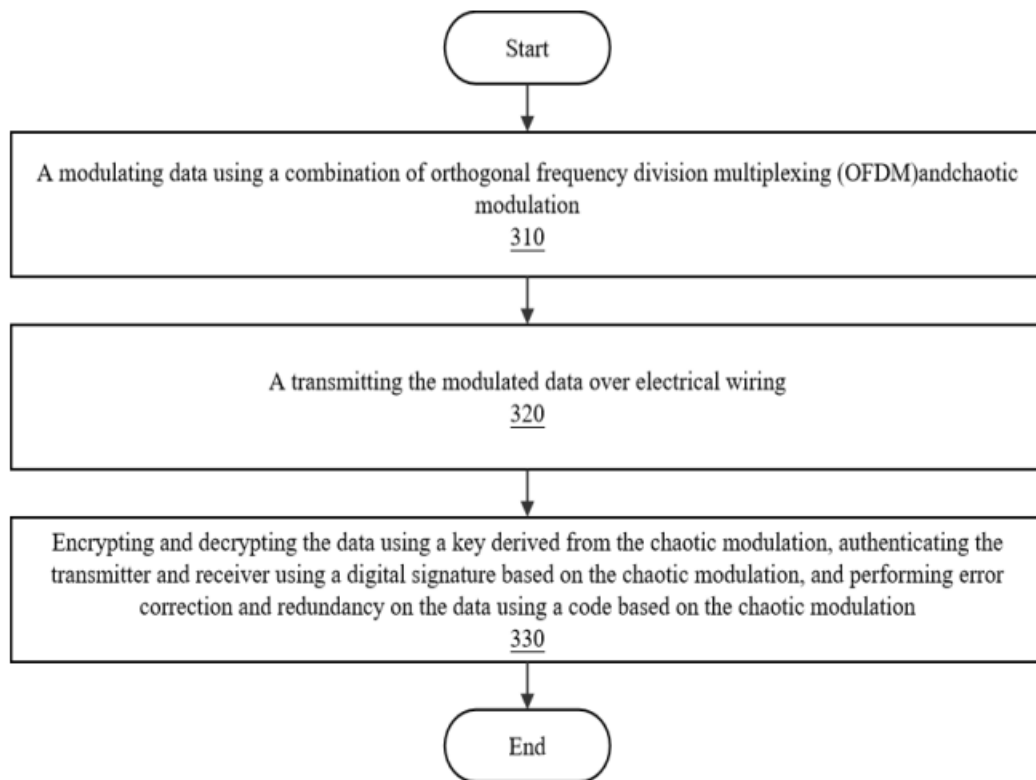
			combination of OFDM and chaotic modulation. The patent emphasizes the system's potential contribution to efficient and sustainable urbanization in line with the objectives of sustainable cities and communities.	
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C. DETAILED DESCRIPTION:

Powerline networking, or powerline communications, is a form of communication that uses electrical wiring to carry both data and alternating current (AC) electrical supply through existing electrical infrastructure. Powerline networking can have a range of uses. In the home it is mostly used for home automation (smart home) and internet access, but it is also used in industrial and commercial settings, for example by utility companies to carry out tasks such as remote meter readings. Data can be transmitted using electrical wires by modulating an electrical signal to represent the data in flow charts. This is commonly done using various encoding schemes, such as amplitude modulation (AM), frequency modulation (FM), or pulse-code modulation (PCM).

Advanced Data Modulation Techniques:

In order to achieve high data rates and low interference over the powerline, the invention uses a novel data modulation technique that combines orthogonal frequency division multiplexing (OFDM) and chaotic modulation. OFDM is a method of dividing a high-bandwidth data signal into multiple lower-bandwidth subcarriers, each modulated by a different frequency. This allows for efficient use of the available spectrum and reduces the effects of multipath fading and noise. Chaotic modulation is a method of encoding data using a chaotic signal, which is a signal that exhibits unpredictable and complex behaviour. This allows for enhanced security and robustness against jamming and eavesdropping, as the chaotic signal is difficult to detect, reconstruct, or predict.



D. RESULTS AND ADVANTAGES:

1. A powerline networking system comprising:
 - a. a transmitter configured to modulate data using a combination of orthogonal frequency division multiplexing (OFDM) and chaotic modulation, and to transmit the modulated data over an electrical wiring; and
 - b. a receiver configured to receive the modulated data from the electrical wiring, and to demodulate the data using a combination of OFDM and chaotic modulation.
2. The powerline networking system as claimed in claim 1, wherein the transmitter and the receiver are further configured to encrypt and decrypt the data using a key derived from the chaotic modulation, wherein the transmitter and the receiver are further configured to authenticate each other using a digital signature based on the chaotic modulation and are further configured to perform error correction and redundancy on the data using a code based on the chaotic modulation.
3. The powerline networking system as claimed in claim 1, wherein the OFDM comprises a plurality of subcarriers, each modulated by a different frequency, and wherein the chaotic modulation comprises a chaotic signal that exhibits unpredictable and complex behaviour, wherein the transmitter is configured to allocate different subcarriers to different data streams based on their priority and bandwidth requirements, and to adjust the power and phase of each subcarrier based on the channel conditions.
4. The powerline networking system as claimed in claim 1, wherein the receiver is configured to detect and synchronize with the subcarriers using a pilot signal embedded in the chaotic signal, and to estimate and compensate for the channel impairments using a channel estimation algorithm based on the chaotic signal.
5. The powerline networking system as claimed in claim 1, wherein the chaotic signal is generated by a nonlinear dynamical system, such as a Lorenz system, a Chua's circuit, or a logistic map.
6. The powerline networking system as claimed in claim 1, wherein the electrical wiring is part of an existing electrical infrastructure that carries both data and alternating current (AC) electrical supply.
7. The powerline networking system as claimed in claim 1, wherein the electrical

infrastructure is used for home automation, internet access, smart grid, or remote metering.

F. EXPANSION: Any variables which are necessary for your invention to be covered?

NA

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION: Is your working prototype or other ready. Provide the images/data of the prototype. If no, how much time is required for the same?

It will take 3-4 months for making the prototype.

G. EXISTING DATA: Any clinical or comparative data necessary enough to support your invention. (Comparative)

NA

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ()	NO (X)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO (X)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ()	NO (X)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO (X)
E. Name of Regulatory body or any other approvals if required.	YES ()	NO (X)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us.

NA

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity).

NA

7. Potential Chances of Commercialization.

Yes

8. List of companies which can be contacted for commercialization along with the website link.

Provide the details of companies.

9. Any basic patent which has been used and we need to pay royalty to them.

NA

10. FILING OPTIONS: Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings (Mandatory to mention).

Complete filing

11. KEYWORDS: Please provide right keywords for searching your invention.

Powerline, Networking System.

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

This is to certify that I (Name of the external person) have no financial assistance in filing any patent form from Lovely Professional University.

We have no objection if Lovely Professional University files any patent with the name of our employee (name of the external person) as co-inventor with (Names of LPU faculty/Staff and all co-inventors) having title as per the patent idea request file (LPU Idea Request I'D....).

Further, our institution will not raise any objections later concerning the filing and commercialization of the said patent.

(Authorised Signatory)